ChArUco calibration target, composite

DATASHEET Rev A · 2026-10-24

ChArUco combines a checkerboard with ArUco markers in the white squares. Markers identify each corner, so the board can be partly outside the image or partly occluded and still contribute corners — which is exactly where lens distortion is strongest. Native in OpenCV (`cv2.aruco.CharucoBoard`), supported by MATLAB and most multi-camera toolchains.

Ordering

Part number	Size (W × H)	Squares (rows × cols)	Square / marker	Dictionary	Corners	Markers
CT-CHA-P-200150-{DICT}	200 × 150 mm	5 × 7	25 / 18 mm	see below	24	17
CT-CHA-P-300200-{DICT}	300 × 200 mm	7 × 10	25 / 18 mm		54	35
CT-CHA-P-400300-{DICT}	400 × 300 mm	9 × 12	30 / 22 mm		88	54
CT-CHA-P-600400-{DICT}	600 × 400 mm	9 × 14	40 / 30 mm		104	63
CT-CHA-P-800600-{DICT}	800 × 600 mm	11 × 15	50 / 37 mm		140	82

`{DICT}`: `6X6` = DICT_6X6_250 (default, most robust) · `5X5` = DICT_5X5_100 · `4X4` = DICT_4X4_50 (decodes at lower resolution, fewer unique IDs) · `AT36` = AprilTag 36h11. Start ID 0, row-major over white squares, OpenCV ≥ 4.6 placement convention (`legacy_pattern: false`). Other grids, dictionaries or start IDs: custom order (CUS).

Pattern area is centred with a white quiet zone ≥ 1 square on all sides; the edge strip with part number, serial and 100 mm scale bar sits outside the quiet zone.

Specifications

Geometry

Feature position tolerance	±0.3 mm + 0.2 mm/m (any corner to nominal grid, at 20 °C)
Square pitch, mean	±0.05 mm
Orthogonality	±0.02°
Marker-to-square ratio	0.74
Edge radius	R6 mm

Physical

Substrate	Aluminium / LDPE / aluminium composite, 6 mm
Weight	6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg)
Pattern	Direct UV print, matte, black on white; optical density ≥ 1.6
Flatness	≤ 0.5 mm per metre, board supported on three points
Rigidity (E·J)	≈ 3800 kN·cm²/m
Mounting	4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture ACC-A-FIX-14 clamps any size

Environment

Operating temperature	–50 to +80 °C
Thermal expansion	≈ 24 µm/(m·°C)
Storage	–20 to +60 °C, dry
Outdoor lifetime	up to 3 years (UV-stable print)
Cleaning	water or ≤ 70 % isopropyl alcohol, soft cloth

Deliverables

Inspection report	`CT-IR-<serial>, per board, dimensional + flatness + visual`
Files	`.opencv.json`, `.kalibr.yaml` (checkerboard mode), `.ros2.yaml`, `.halcon.descr`, `.dxf`
Documents	this datasheet, CT-UM-GEN user manual, CT-UM-CHA-P quick start
Lead time	Made to order · ships in 2–3 weeks
Options	`ILAC-MRA certificate issued by an accredited third-party laboratory, available as SVC-X-CERT-ILAC (+1-`

Compliance and customs

RoHS-compliant materials. HS 9031 80. Not a toy; not for food contact.

Revision history

Rev	Date	Change
A	2026-10	First issue, planning values

Specifications subject to change; current revision at calib.tools/targets/charuco-composite/